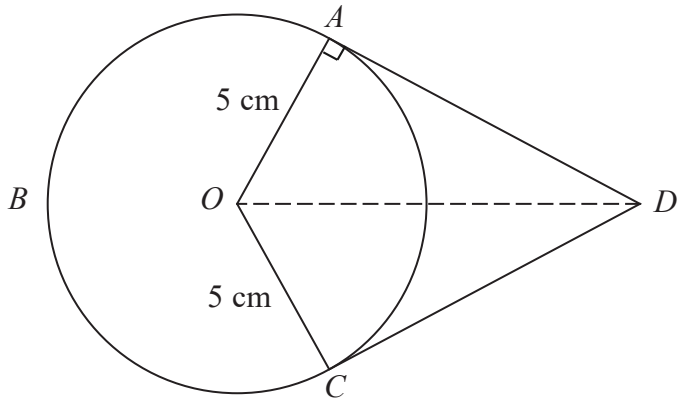
18



A, *B* and *C* are points on a circle of radius 5 cm, centre *O*.
DA and *DC* are tangents to the circle.
DO = 9 cm

Work out the length of arc *ABC*.
Give your answer correct to 3 significant figures.

Radius OA is perpendicular to tangent AD.

1 mark
Recognise angle OAD = 90°

Let θ = angle AOD. $\cos\theta = \frac{5}{9}$

1 mark
Adjacent ÷ hypotenuse

$\theta = \cos^{-1}(5 \div 9) = 56.251011...°$

1 mark
Use inverse cosine

Triangles OAD and OCD are congruent (RHS).

Minor angle AOC = 2θ; major angle = $360° - 2\theta$.

Arc ABC = $\frac{360 - 2\theta}{360} \times 2\pi \times 5$

1 mark
Use the major angle as a fraction of 360°

Arc ABC = 21.598272... cm ≈ 21.6 cm

1 mark
Round to 3 significant figures

21.6 cm

(Total for Question 18 is 5 marks)